FELICITYBENCH: Pragmatic (In)felicity Sensitivity in LLMs

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Abstract

We introduce **FELICITYBENCH**, a benchmark for evaluating large language models' pragmatic competence with respect to felicity conditions, the contextual and social requirements that must be satisfied for a speech act to be successfully performed. The benchmark comprises 50 utterances, each paired with three contexts rendering the utterance felicitous, infelicitous, or an edge case, across the five speech act types as defined by J.L. Austin and John Searle. We evaluate six open-weight models on two tasks: a three-way felicity recognition task and a behavioral response task in which models interact with infelicitous utterances in a naturalistic conversational setting. All models exceed chance on the recognition task, though performance varies substantially across models and is much worse at handling edge cases. Behavioral response strategies vary dramatically across models and do not correlate with recognition accuracy. We argue that this mismatch has direct consequences for how pragmatic competence is assessed in LLMs, highlighting a key way in which their mode of cognition and communication differs significantly from our own.

1 Introduction

"You have a mild bacterial infection that should clear up with antibiotics."

How would a human language user go about computing the meaning of the above utterance? Linguists and psychologists agree that the cognitive process of parsing language input happens at multiple different levels. The sonic (or visual, in the case of written or signed language) stream is mapped onto a finite set of phonemic representations, the sequence of which is segmented, which each segment mapped onto a *lexeme* (more or less, a word). The often multiplicitous and complex meanings of these words are combined via a compositional process, roughly tracking the abstract hierarchical structure by which these words are organized in the stream, in order to compute the meaning of the entire expression. However, the process does not stop there: the meaning of the expression generated at the end of this process is still impoverished in several ways. As it turns out, the meanings of the linguistic forms we produce are deeply dependent upon the *context* in which we use them. The study of the mechanisms by which context modulates meaning is known as *pragmatics*.

Returning to the sentence above, consider the following set of contexts:

1. A doctor has examined you and says this after reviewing your lab results.
2. A stranger approaches you while you are sitting at a park and says this to you.
3. Your roommate returns home, and upon noticing that you are sick, tells you this.

The meaning (and indeed, the *meaningfulness*) of the utterance changes dramatically depending on the differing contexts. In context 1, the utterance is a diagnosis—in saying this, the doctor is providing a professional, trustworthy perspective on the state of your health, as well as providing a course of action informed by their medical expertise. In context 2, the utterance may as well be entirely meaningless—a stranger approaching you and attempting to perform a diagnosis is entirely inappropriate. Even if the stranger had medical expertise, they would still require familiarity with you and your health in order to diagnose. In

context 3, things are a bit more complicated; unless your roommate just so happened to be an infectious disease specialist who could identify the severity and source of an illness upon first glance, the utterance is somewhat problematic given in any authoritative manner. At the same time, it is somewhat reasonably interpretable as a good-faith guess, though perhaps one given in an overly assertive style.

Felicity conditions, introduced by philosopher of language J.L. Austin, identify the contextual and social requirements that must be satisfied for a speech act to be successfully performed or a sentence to carry out its full communicative force. They are inextricably linked to the knowledge states of each interlocutor and the relationships between them, and the seamless fulfillment of these conditions in everyday interactions demonstrates language users' sophisticated understanding of social-linguistic conventions. In these terms, context 1 fulfills the utterance's felicity conditions, rendering it *felicitous*. Context 2, in turn, violates the utterance's felicity conditions, rendering it *infelicitous*. Of course, felicity is not strictly binary: contexts like 3 exemplify the sorts of scenarios that might not render an utterance straightforwardly felicitous or infelicitous, but rather constitute a sort of edge case.

Much work has gone into evaluating LLMs' capacity to function as pragmatic speakers, specifically in their production of context-sensitive and felicitous responses when prompted. Less work has gone into evaluating LLMs' *understanding* of felicity conditions as a distinct factor influencing successful communication. In this realm, we particularly seek to address two areas: **(1) Recognition**: Given a particular conversational context, do LLMs recognize the conditions that render an utterance felicitous or infelicitous? **(2) Response**: How do LLMs typically respond to infelicitous utterances of various forms? Do they ignore the violation, adapt the context to accommodate the utterance, or explicitly flag the infelicity?

Through behavioral experiments assessing LLMs' recognition of felicity conditions and their responses to pragmatic infelicity, we determine whether these models demonstrate explicit awareness of such conditions and how this awareness influences their behavior, if it does. In order to run these experiments, we have developed **FELICITYBENCH**, a dataset of 50 utterances, each with 3 contexts, with one rendering the paired utterance felicitous, one rendering it infelicitous, and one constituting an edge case. The scenarios cover the five speech acts types in the Austin/Searle taxonomy: assertives, directives, commissives, expressives, and declaratives.

As felicity conditions are grounded in a theory of *human* communication, it is nontrivial to assume that they operate identically in non-human agents; as such, our project will also shed light on the extent to which linguistic pragmatics may struggle to fully account for the communicative pressures shaping linguistic interaction involving cognitive aliens.

2 Related Work

We situate our work at the intersection of **(1)** established work in linguistic theory and the philosophy of language and **(2)** the rapidly emerging research direction applying social scientific methods to study machine intelligence.

2.1 Speech act theory and felicity conditions.

Austin (1962) speech act theory and Searle's (1969) systematization establish that utterances perform actions (asserting, promising, ordering) subject to felicity conditions: contextual prerequisites that need to be met for the act to succeed. Performative utterances, as opposed to constatives which describe states of affairs and can be evaluated as true or false, enact an action upon utterance and can only be evaluated as felicitous or infelicitous. Felicity thus captures speakers' intuitions about whether an utterance occurs appropriately in context. Searle (1969) classifies performative speech acts into five types (assertives, directives, commissives, expressives, declaratives), which is the framework adopted to construct the dataset for this study to test whether models recognize when a context provides the necessary conditions for an utterance to be felicitous:

- Assertives are utterances that describe states of affairs and publicly commit the speaker to a proposition.
- Directives are utterances that attempt to achieve a specific result and get the listener to take a specific action, such as ordering and requesting.
- Commissives, such as promising, commit the speaker to a certain future course of action.
- Expressives primarily convey the speaker's attitude or psychological state regarding a situation.
- Declaratives produce a change in the reality of status of an object or situation upon its immediate utterance, as in adjourning a meeting.

Austin (1962) further distinguishes two kinds of infelicity: misfires, where the speech act fails to be performed either because the relevant convention does not exist or is not invoked by the appropriate persons (misinvocations), or because the procedure exists but is badly executed (misexecutions); and abuses, where the act is performed but the participants lack the required thoughts, intentions, or feelings, as in insincere promises or apologies.

A related source of infelicity is presupposition failure, where a sentence's presupposition, i.e., its underlying assumptions, rendering it inappropriate or infelicitous relative to the common ground of the conversation (Stalnaker, 1974). Gostev (n.d.) measures whether models detect broken premises and refuse to engage with them, rather than confidently continuing with invalid assumptions, which can be understood as a test of sensitivity to presupposition failure and infelicitous input. It is worth noting that while some performatives bear characteristic morphosyntactic features (first person, present indicative, active), the means by which a particular speech act can be performed are highly varied. A performative verb such as "promise" only makes explicit an action that can be performed in many other ways. Conversely, utterances sharing the same sentence force, which is a conventional aspect of language and part of its grammar, can be used to perform different speech acts depending on context.

2.2 Evaluation of LLM Pragmatic Competence

Existing benchmarks evaluating LLMs' pragmatic competence show that despite their ability to produce linguistically fluent output, LLMs often fall short of genuine pragmatic competence compared to human speakers. Sravanthi et al. (2024) offer a comprehensive evaluation across 14 tasks including implicature, presupposition, reference, and deixis, finding a 15–25% gap between LLMs and humans on presupposition and figurative language tasks. This study and Kabbara & Cheung (2022) both suggest that LLMs' understanding of language may be more significantly influenced by the presence of specific structural and lexical cues than by inherent pragmatic logic. This raises the question of whether LLMs' apparent sensitivity to felicity conditions similarly reflects surface pattern matching rather than genuine pragmatic reasoning. Pan & Kehler (2025) investigate conversational eliciture, a type of pragmatic inference where the felicity of the utterance is not at stake, finding that larger models can use pragmatic inference to guide downstream syntactic processing in ways that pattern with human behavior, whereas smaller models struggle to draw elicitures at all. This scale-dependent result motivates testing whether the more foundational aspect of pragmatic competence—sensitivity to felicity conditions—shows a similar gradient across models.

Jian (2024) examine LLMs as pragmatic speakers using the Rational Speech Act (RSA) framework (Goodman & Frank, 2016) in a reference game paradigm, focusing on production rather than comprehension. RSA treats pragmatic reasoning as a recursive inference process grounded in cooperative rationality, providing a theoretical basis for expecting competent models to reason about speaker intentions and contextual appropriateness. While LLM performance scores were found to correlate positively with RSA predictions, there is insufficient evidence to conclude that models behave as genuine pragmatic speakers. Ma et al. (2025) corroborate this picture in a broader survey of pragmatic evaluation resources in NLP, finding that LLMs still lack robustness in pragmatic inference, failing to recognize

indirect requests or subtle shifts in social meaning, and noting that existing benchmarks are heavily dominated by English-focused datasets, limiting cross-linguistic and cultural generalizability.

Speech act theory captures an essential dimension of meaning that emerges from interaction rather than isolated sentences, and has increasingly been applied to the analysis and evaluation of LLMs. Li (2025) draws on the speech act distinction between locution, illocution, and perlocution to reframe hallucination in LLMs as a pragmatic failure: an infelicitous performative in which an assertion or directive is issued without adequate authority, evidence, or situational fit. The proposed conceptual solution is a context-sensitive, pragmatic-gating layer that licenses outputs only when the relevant felicity conditions are met. Ma et al. (2025) similarly argue that speech act theory provides opportunities for pragmatics to inform LLM alignment. Together, these accounts position felicity conditions as theoretically central to understanding LLM behavior. Yet, whether LLMs are actually sensitive to such conditions remains underexplored empirically. Our work directly addresses this gap, using Searle's taxonomy as an evaluation framework to test whether models recognize when contextual conditions license a given speech act as appropriate.

2.3 Our contribution

While other work has characterized other aspects of LLMs' pragmatic competence, little work has gone into specifically assessing their handling of pragmatic (in)felicity. Our work provides a benchmark by which LLMs may be evaluated for their ability to both recognize and effectively respond to cases of infelicity. LLMs' ability to perform both of these skills becomes increasingly crucial as they continue to increase their participation in human speech communities—especially given widely-noted patterns of sycophancy exhibited by many LLMs as a result of their post-training, their ability to function as independent communicative agents critically depends on their ability to effectively handle situations in which their inputs are inappropriate given their contexts.

3 Methods

We conduct two experiments. The first is a **recognition** task, asking the model to assess whether a certain utterance is felicitous, infelicitous, or an edge case, based on a given context. The models are scored based on their accuracy. The second task is a **response** task, asking the model to respond to an utterance given an infelicitous context. These responses are coded as one of {IGNORE, ACCOMMODATE, FLAG, REPAIR}, representing four classes of possible model behaviors. All experiments were run using the **FELICITYBENCH** dataset.

3.1 Dataset

The **FELICITYBENCH** dataset consists of 50 distinct utterances, designed to be diverse and comprehensive across the five types of speech act as identified by Austin and Searle. The 50 are evenly split across these types: there are 10 assertives, 10 directives, 10 commissives, 10 expressives, and 10 declaratives. Each of the 50 utterance is tied to 3 contexts, one which renders the utterance felicitous (meeting all contextual requirements to have its full communicative or performative force), one which renders the utterance infelicitous (being entirely inappropriate to utter given the context), and one constituting an edge case (questionable or slightly inappropriate on its own but perhaps somewhat acceptable with additional context).

The dataset was created by carefully hand-designing "seed" datapoints manually, with special focus placed on crafting edge contexts that were plainly neither felicitous nor infelicitous along with clearly felicitous and infelicitous contexts, and with intentional diversity, both in spanning across the five utterance types and also across varied registers of formality and social context (with differing relations between interlocutors, with some more focused on things like familial relations, others more related to institutional roles, and still others focused on plain, relatively social-context-free access to information. The "seed" datapoints were structured so as to fit cleanly with the experimental pipelines and then

were provided to Claude Sonnet 4.6, which was tasked with generating more datapoints with a similar diversity, incorporating the original seed datapoints to reach 10 of each speech act category. This dataset was continually refined until it reached a sufficient degree of naturalism.

The design of the dataset is such that all utterances and contexts are syntactically and semantically well-formed—they differ only in the *pragmatic* factors that they require/induce. Particular focus was put on making sure there were no structural "tells" allowing LLMs to learn patterns of pragmatic felicity as a correlation with specific syntactic or semantic information in the datapoints. As a result, it allows for the isolation of LLMs' pragmatic sensitivities as the object of study.

3.2 Model Architecture

The experiments were run on 6 different LLMs developed by 4 different providers and spanning three different size ranges (by parameter count). The models tested were `gemma-3-27b-it` by Google, `Qwen2.5-72B-Instruct` and `Qwen2.5-32B-Instruct` by Qwen, `Llama-3.1-70B-Instruct` and `Llama-3.1-8B-Instruct` by Meta, and `Mistral-7B-Instruct-v0.3` by Mistral. The range of models allows us to evaluate if different architectures have differing performance; the inclusion of `Qwen2.5` and `Llama-3.1` at a larger and a smaller size each also allows us to see the effects of model size on performance, within an otherwise similar architecture.

3.3 Experimental Setup

Models were sourced from HuggingFace; all experiments were run on Modal GPUs (NVIDIA H100 and A10G GPUs, depending on the size of the model being run). Each model was loaded in sequence and presented with the two experiments:

3.3.1 Experiment 1: Recognition

Experiment 1 measured the LLMs' ability to classify a certain utterance as felicitous, infelicitous, or an edge case, depending on the context provided. The experiment was run with two different conditions, an "expert condition" and a "non-expert condition", which differed on the basis of the system prompts provided. The exact prompts can be found in the Appendix, but the expert condition began with the line "You are a linguistics expert specializing in pragmatics and speech act theory", and continues on to give definitions of "felicitous" and "infelicitous", thus explicitly prompting the model to utilize specialist knowledge about pragmatics as an academic discipline to make its assessments. The non-expert condition removes this entirely, avoiding any and all technical language and simply asking the models to classify the utterances as "appropriate", "inappropriate", or "edge" depending on the context provided.

3.3.2 Experiment 2: Response

Experiment 2 measured the LLMs' responses to utterances given contexts that render them infelicitous. Unlike Experiment 1, which involved direct classification, Experiment 2 allowed for free response from the model, producing a wide range of differing responses. The responses were then coded using the LLM-as-a-Judge technique. The judge model was `Qwen2.5-72B-Instruct` in all cases, which is subsequently prompted to classify each response as an instance of one of the four following response types: **(1)** IGNORE: The model responds as if the utterance were perfectly normal and appropriate, with no acknowledgment that anything is wrong or unusual. **(2)** ACCOMMODATE: The model adjusts the situation or interpretation to make the utterance seem plausible (e.g., by inventing a backstory, assuming the speaker has hidden qualifications, or reframing the scenario). **(3)** FLAG: The model explicitly points out that the utterance is inappropriate, out of place, unusual, or that the speaker lacks the authority/standing/basis to say it. **(4)** REPAIR: The model suggests a correction, alternative, or redirect that would make the interaction more

appropriate (e.g., suggesting the person consult an actual expert). Each model's distribution of response types, separated by differing speech act types, are reported.

3.3.3 Extension from hypogenic.ai Repository

The experiments were based on the AI-generated GitHub repository from hypogenic.ai, but with major augmentations made so as to strengthen the naturalism and significance of the results. The most significant change was the overhaul of the AI-generated dataset and its replacement with one that incorporates a three-way split, with contexts constituting edge cases alongside those rendering the utterances felicitous or infelicitous. This required major structural changes to the scripts. Another change was the addition of the non-expert condition to the recognition task. The AI-generated experiment directly prompted the LLMs to assume the role of linguistics experts and thus draw on a specialized knowledge base. This posed a major theoretical problem to the naturalism of the results of the study—felicity assessment itself is an analytical framework within a scientific discipline, not a direct reflection of humans' unconscious judgments about the appropriateness of utterances, which is the object of study, both of this analysis and of felicity assessment itself. As such, we sought to compare this "expert condition", retaining this original prompting with a "non-expert condition" with no reference to any specialized knowledge, to better reflect the naturalistic experience of interpreting utterances, as detailed above. Other changes mainly had to do with adapting a framework built for enterprise models (in the AI-generated repository) to work with our set of accessible open-source models.

4 Results

Model	Assertives	Directives	Commissives	Expressives	Declaratives
`Qwen2.5-72B-Instruct`	$0.6\overline{6}$	$0.4\overline{3}$	$0.5\overline{3}$	0.6	$0.6\overline{6}$
`Llama-3.1-70B-Instruct`	0.6	$0.5\overline{6}$	0.7	$0.6\overline{3}$	0.7
`Qwen2.5-32B-Instruct`	0.6	0.4	0.5	0.5	$0.6\overline{3}$
`gemma-3-27b-it`	$0.6\overline{6}$	$0.5\overline{6}$	$0.6\overline{3}$	0.7	0.7
`Llama-3.1-8B-Instruct`	$0.4\overline{3}$	0.4	$0.3\overline{3}$	$0.5\overline{3}$	0.5
`Mistral-7B-Instruct-v0.3`	0.5	$0.3\overline{6}$	$0.3\overline{6}$	0.4	$0.6\overline{3}$

Table 1: Recognition accuracy by speech act type.

Model	Felicitous	Edge	Infelicitous
`Qwen2.5-72B-Instruct`	0.88	0.24	0.62
`Llama-3.1-70B-Instruct`	0.88	0.22	0.82
`Qwen2.5-32B-Instruct`	0.92	0.20	0.46
`gemma-3-27b-it`	0.92	0.36	0.68
`Llama-3.1-8B-Instruct`	0.70	0.36	0.26
`Mistral-7B-Instruct-v0.3`	0.86	0.28	0.22

Table 2: Recognition accuracy by felicity condition.

4.1 Experiment 1: Felicity Condition Recognition

Overall three-way classification accuracy (felicitous / edge / infelicitous) is reported in Table 2. All six models exceed the chance baseline of $p = 0.\overline{3}$. `gemma-3-27b-it` achieves the highest accuracy (0.653; Cohen's $h = 0.652$), followed by `Llama-3.1-70B-Instruct` (0.640; $h = 0.624$), `Qwen2.5-72B-Instruct` (0.567; $h = 0.501$), `Qwen2.5-32B-Instruct` (0.527; $h = 0.393$), `Mistral-7B-Instruct-v0.3` (0.453; $h = 0.246$), and `Llama-3.1-8B-Instruct` (0.440; $h = 0.220$). A McNemar's test between the best- and worst-performing models is significant ($\chi^2 = 18.48$, $p < 0.001$).

Model	χ^2	p	df	Note
`Qwen2.5-72B-Instruct`	4.82	0.307	4	
`Llama-3.1-70B-Instruct`	1.85	0.763	4	
`Qwen2.5-32B-Instruct`	4.12	0.390	4	
`gemma-3-27b-it`	1.65	0.800	4	
`Llama-3.1-8B-Instruct`	3.08	0.544	4	
`Mistral-7B-Instruct-v0.3`	6.35	0.175	4	
`gemma-3-27b-it` vs. `Llama-3.1-8B-Instruct`	18.48	<0.001	1	McNemar

Table 3: Chi-squared tests for speech act type effect on recognition accuracy (rows 1–6), and McNemar's test between best- and worst-performing models (row 7).

Model	Cohen's h
`Qwen2.5-72B-Instruct`	0.501
`Llama-3.1-70B-Instruct`	0.624
`Qwen2.5-32B-Instruct`	0.393
`gemma-3-27b-it`	0.652
`Llama-3.1-8B-Instruct`	0.220
`Mistral-7B-Instruct-v0.3`	0.246

Table 4: Effect sizes (Cohen's h) relative to chance ($p = 0.\overline{3}$) for each model on the recognition task.

Accuracy by felicity condition. Table 2 reports accuracy broken down by felicity condition. Felicitous items yield the highest accuracy across all models (range: 0.70–0.92), followed by infelicitous items (range: 0.22–0.82), with edge cases consistently the lowest (range: 0.20–0.36).

Accuracy by speech act type. Table 1 reports accuracy by speech act type. Chi-squared tests indicate no significant effect of speech act type on recognition accuracy for any model (all $p > 0.17$, all $\chi^2 < 6.35$, $\text{df} = 4$).

Model	Ignore	Accommodate	Flag	Repair
`Qwen2.5-72B-Instruct`	0.20	0.30	0.08	0.42
`Llama-3.1-70B-Instruct`	0.12	0.12	0.58	0.18
`Qwen2.5-32B-Instruct`	0.00	0.30	0.10	0.60
`gemma-3-27b-it`	0.00	0.58	0.16	0.26
`Llama-3.1-8B-Instruct`	0.06	0.02	0.16	0.76
`Mistral-7B-Instruct-v0.3`	0.00	0.50	0.34	0.16

Table 5: Distribution of response strategies (proportion) for each model when presented with infelicitous utterances.

4.2 Experiment 2: Behavioral Response to Infelicity

Response strategy distribution. Table 5 reports the distribution of response strategies by model. Response category distribution differs significantly across models ($\chi^2 = 140.59$, $p < 0.001$, $\text{df} = 15$, Cramér's $V = 0.395$). Accommodation is the dominant strategy for `gemma-3-27b-it` (0.58) and `Mistral-7B-Instruct-v0.3` (0.5); repair for `Llama-3.1-8B-Instruct` (0.76) and `Qwen2.5-32B-Instruct` (0.6); and flagging for `Llama-3.1-70B-Instruct` (0.58). Ignore rates are low across models, with `Qwen2.5-72B-Instruct` the highest at 0.20.

Comparison	χ^2	p	df	Cramér's V
Response category × Model	140.59	<0.001	15	0.395
Response category × Speech act	13.73	0.319	12	0.123

Table 6: Chi-squared tests for the effect of model and speech act type on response strategy distribution.

Model	Flag Rate
`Llama-3.1-70B-Instruct`	0.580
`Mistral-7B-Instruct-v0.3`	0.340
`gemma-3-27b-it`	0.160
`Llama-3.1-8B-Instruct`	0.160
`Qwen2.5-32B-Instruct`	0.100
`Qwen2.5-72B-Instruct`	0.080

Table 7: Proportion of responses in which each model explicitly flags infelicity, sorted in descending order.

Flag rate. Flag rates by model are reported in Table 7. `Llama-3.1-70B-Instruct` produces the highest flag rate (0.58), followed by `Mistral-7B-Instruct-v0.3` (0.34), `gemma-3-27b-it` and `Llama-3.1-8B-Instruct` (0.16 each), `Qwen2.5-32B-Instruct` (0.1), and `Qwen2.5-72B-Instruct` (0.08).

Response strategy by speech act type. Response category distribution does not differ significantly by speech act type ($\chi^2 = 13.73$, $p = 0.319$, df $= 12$, Cramér's $V = 0.123$).

4.3 Relationship Between Recognition and Behavioral Response

Table 8 reports recognition accuracy alongside flag rate for each model. The Spearman correlation between the two measures is $\rho = -0.14$. `gemma-3-27b-it`, the highest-accuracy recognizer (0.653), produces a flag rate of 0.16, equal to that of `Llama-3.1-8B-Instruct` (0.44). `Llama-3.1-70B-Instruct` pairs the second-highest recognition accuracy (0.64) with the highest flag rate (0.58). `Qwen2.5-72B-Instruct` achieves moderate recognition accuracy (0.567) alongside the lowest flag rate (0.08).

5 Discussion

We see that by and large, LLMs' ability to recognize infelicity does not readily translate into behavior utilizing this recognition to explicitly flag instances of pragmatic infelicity when they are encountered with them. This is likely due to a confluence of factors.

To begin, Experiment 1 and Experiment 2 represent two substantially different sorts of pragmatic tasks, which can be broadly thought of as production and comprehension, respectively. By our results, we can observe that the LLMs seem to have more robust *productive* capacity than *comprehensive* capacity when it comes to infelicity: they can classify utterance-context pairs at degrees substantially higher than chance, but almost never explicitly flag infelicitous utterances when they are presented with them. While this may be a fact related to the model architecture, a very likely hypothesis is that this is a result of models' post-training. Because the two experiments constitute such different tasks, responding to each carries a different sort of communicative weight; while Experiment 1 is a classification task simply requiring the model to generate a single word per trial and thus has no real extra communicative significance, Experiment 2 essentially requires the model to be directly confrontational to a user. As has been noted, many models develop a characteristic *sycophancy* through their post-training process, learning to bias their outputs towards maximal friendliness and cooperativity, to the point of becoming somewhat nonconfrontational when faced with

Model	Recognition Accuracy	Flag Rate
Qwen2.5-72B-Instruct	0.567	0.080
Llama-3.1-70B-Instruct	0.640	0.580
Qwen2.5-32B-Instruct	0.527	0.100
gemma-3-27b-it	0.653	0.160
Llama-3.1-8B-Instruct	0.440	0.160
Mistral-7B-Instruct-v0.3	0.453	0.340

Table 8: Recognition accuracy (Experiment 1) and flag rate (Experiment 2) by model. Spearman $\rho = -0.14$.

infelicitous situations. At the scale of individual users, this orientation likely makes LLM interaction more approachable, significantly lessening pressures of social performance that can make human-to-human interactions affectively taxing. From a bird's eye view, however, this means that the human-to-LLM interaction paradigm is one that bears a significant difference from the human-to-human interactions that characterize the rest of human social life, a fact that is not always consciously recognized and accommodated by users.

Llama-3.1-70B-Instruct is a notable outlier in all this, with a much higher flag rate than any of the other models in question, alongside high recognition accuracy. Interestingly, we see that its smaller cousin, Llama-3.1-8B-Instruct, does not demonstrate the same eagerness to flag as the larger model does (though Mistral-7B-Instruct-v0.3 seems to be potentially similar). While the exact reason for this unique behavior is unclear, and likely has to do with the particular post-training it received, the fact that such an outlier exists goes to show that the degree of sycophancy exhibited by the other models is not intrinsic to the architecture.

Another interesting pattern highlighted by this work is that while LLMs overall seem to enjoy a high degree of accuracy in the recognition of felicitous and infelicitous utterance-context pairs, their performance suffers dramatically when it comes to classifying edge cases. This is a very significant finding and has implications for further refinement of pragmatics benchmarks—since felicity and other pragmatic measures are not necessarily binaristic in the real world, treating them as such in benchmarks may lead to the overstatement of the evaluated LLMs' performance and capabilities. Relatedly, this "U-shaped" performance also may indicate that LLMs' internal representations of pragmatic factors are categorical in nature, rather than mapping onto the more complicated and graded way they appear to manifest in human communication.

Due to hardware constraints, we were unable to evaluate larger models we hoped to like Qwen3.5-397B-A17B and DeepSeek-R1; future research could address how such models perform compared to those in the present study. With the latter model especially, it would be interesting to see if reasoning models see a boost in performance, especially if they fare better at recognizing edge cases.

6 Conclusion

We introduced **FELICITYBENCH**, a benchmark for evaluating LLMs' recognition of and behavioral response to felicity condition violations across the five speech act types as stipulated by Austin and Searle, and reported results for six open-weight models. We find that while models are able to successfully recognize straightforwardly felicitous and infelicitous utterances, they struggle with edge cases, and furthermore, their ability to recognize (in)felicity does not translate into an ability to explicitly flag it when it occurs. Llama-3.1-8B-Instruct bucks the trend in a very significant way, demonstrating that robust infelicity flagging is not incompatible with the LLM's basic architecture, even with instruction tuning. These results have direct implications for how pragmatic competence ought to be evaluated in LLMs; future work should be careful to consider intermediary/ambiguous cases to test LLMs' discriminatory capabilities, and investigate the extent to which sycophantic tendencies might interfere with LLMs' ability to communicate effectively when faced with utterances inappropriate for the contexts in which they are encountered.

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A Appendix

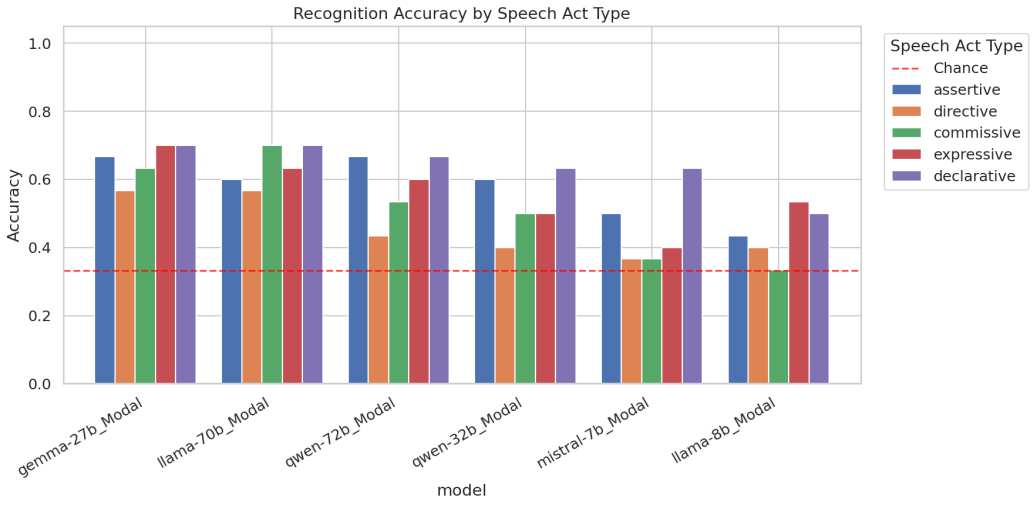


Figure 1: Recognition by Speech Act.

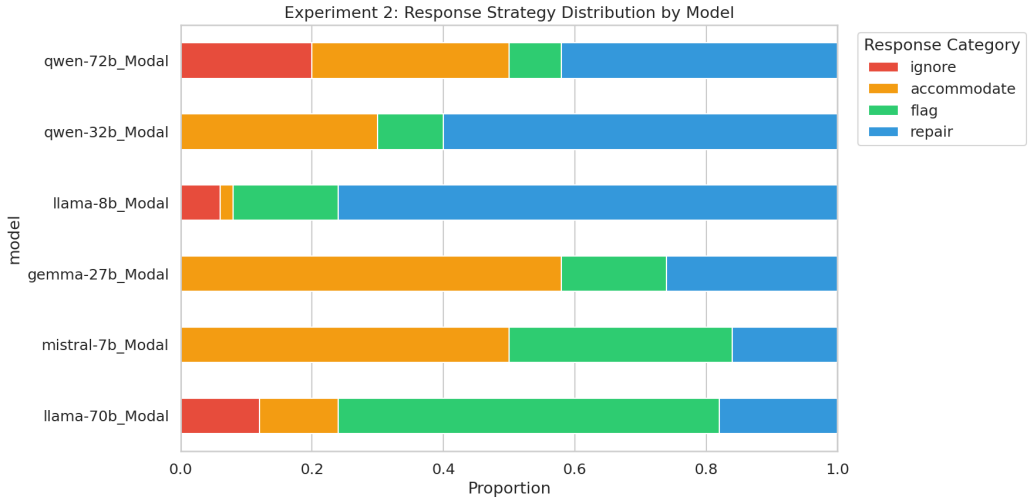


Figure 2: Response Distribution.

A Prompt Templates

All prompts were constructed programmatically by filling the templates below with scenario-specific fields drawn from **FELICITYBENCH**. Placeholder fields are indicated in *italics*.

A.1 Experiment 1: Recognition Task

The following prompt was used for expert condition recognition trials. The system prompt was: *"You are a linguistics expert specializing in pragmatics and speech act theory. Answer the question precisely and concisely."*

> Consider the following conversational scenario:
>
> Context: *[scenario context]*
>
> Someone says: "*[utterance]*"
>
> In pragmatics, a speech act is 'felicitous' when all the appropriate conditions for it to be performed successfully are met (e.g., the speaker has the authority, knowledge, ability, or standing to make such a statement in this context). A speech act is 'infelicitous' when these conditions are violated.
>
> A speech act is 'edge' when it is slightly infelicitous, but the speaker has some authority or justification for making the statement.
>
> Is this utterance felicitous, edge, or infelicitous in the given context?
>
> Answer with exactly one word: 'felicitous', 'edge', or 'infelicitous'.

An alternate version removed all specialist knowledge to yield the non-expert condition. The system prompt was simplified to: *"Answer the question precisely and concisely."*

> Consider the following conversational scenario:
>
> Context: *[scenario context]*
>
> Someone says: "*[utterance]*"
>
> Is this utterance appropriate, inappropriate, or edge (slightly inappropriate but can be made appropriate with additional information) in the given context?
>
> Answer with exactly one word: 'appropriate', 'edge', or 'inappropriate'.

A.2 Experiment 2: Behavioral Response Task

The following prompt was used for all behavioral response trials. The system prompt was: *"You are a helpful participant in a conversation. Respond naturally and honestly."*

> You are a participant in the following conversation.
>
> Context: *[scenario context]*
>
> The other person says to you: "*[utterance]*"
>
> How do you respond?

A.3 Judge Model Prompt

The following prompt was used to classify model responses in Experiment 2. Classifications were produced by `Qwen2.5-72B-Instruct`, with system prompt: *"You are an expert in pragmatics and speech act theory. Classify the response accurately."*

> You are analyzing how a language model responded to a pragmatically infelicitous utterance.
>
> BACKGROUND: In this scenario, someone makes an utterance that violates felicity conditions:
>
> Context: *[scenario context]*
> Utterance: "*[utterance]*"
> Why it's infelicitous: *[violation description]*
>
> MODEL'S RESPONSE: "*[model response]*"
>
> Classify the model's response into EXACTLY ONE of these categories:
>
> **ignore**: The model responds as if the utterance were perfectly normal and appropriate, with no acknowledgment that anything is wrong or unusual.
>
> **accommodate**: The model adjusts the situation or interpretation to make the utterance seem plausible (e.g., by inventing a backstory, assuming the speaker has hidden qualifications, or reframing the scenario).
>
> **flag**: The model explicitly points out that the utterance is inappropriate, out of place, unusual, or that the speaker lacks the authority, standing, or basis to say it.
>
> **repair**: The model suggests a correction, alternative, or redirect that would make the interaction more appropriate (e.g., suggesting the person consult an actual expert).
>
> Respond with ONLY the category name (IGNORE, ACCOMMODATE, FLAG, or REPAIR) on the first line, followed by a brief one-sentence justification on the second line.

B Dataset Samples

The following are representative examples from **FELICITYBENCH**, illustrating the three-context structure and violation annotation scheme across three speech act types. Each item comprises a single utterance paired with a felicitous, edge, and infelicitous context, along with a violation type and description applicable to the edge and infelicitous conditions.

Sample 1 *Assertive*

> **Utterance:** "You have a mild bacterial infection that should clear up with antibiotics."
>
> **Felicitous context:** A doctor has examined you, the patient, and says this after reviewing your lab results.
>
> **Edge context:** Your roommate returns home, and upon noticing that you are sick, tells you this.
>
> **Infelicitous context:** A stranger approaches you while you are sitting at a park and says this to you.
>
> *Violation type:* Preparatory. *Violation description:* Speaker lacks full diagnostic authority.

Sample 2 *Directive*

Utterance: "I'm telling you, you have to stop drinking the tap water — it's not safe!"

Felicitous context: Your brother says this to you after you take a sip of tap water from a sink in an old building with lead pipes.

Edge context: You are at a bar and just drank a glass of water given to you by a bartender, who filled the glass out of your view but within your friend's view. Your friend says this to you after you drank the glass of water.

Infelicitous context: Another hiker says this to you as they pass you in the middle of the woods while you are hiking up a mountain trail.

Violation type: Preparatory. *Violation description:* Improper physical situation to make such a directive.

Sample 3 *Commissive*

Utterance: "If you don't stop cheating, I'll have you suspended from school."

Felicitous context: Your high school's principal says this to you, a student, after you have been repeatedly caught cheating on exams in several of your classes.

Edge context: One of your classmates, who sees you sneaking a glance at their answer paper during a test, says this to you after the test is over.

Infelicitous context: A cashier at the grocery store says this to you after overhearing you telling your friend that you looked up the answers to your homework.

Violation type: Preparatory. *Violation description:* Speaker lacks disciplinary authority.

C Experiment 2: Sample Model Responses

The following are representative examples of model responses from Experiment 2, illustrating each of the four response categories — accommodate, flag, repair, and ignore — as classified by the judge model (`Qwen2.5-72B-Instruct`). Each example includes the infelicitous context, the utterance, the model's response, and the judge's classification with justification.

Sample 1 *Accommodate* — `gemma-3-27b-it`

Speech act type: Directive

Utterance: "I'm telling you, you have to stop drinking the tap water — it's not safe!"

Context: Another hiker says this to you as they pass you in the middle of the woods as you are hiking up a mountain trail.

Violation type: Preparatory. *Violation description:* Improper physical situation to make such a directive.

Model response: "Oh, really? That's good to know, thanks for the heads up! I've been using my filter, but I appreciate you mentioning it. Do you

know why it's not safe? Is it something specific to this area, or generally? I always try to be careful, but it's good to get info from people actually on the trail. I hope you have a good hike!"

Classification: ACCOMMODATE. The model accepts the unsolicited advice as helpful information and seeks to establish a reasonable context for the utterance by inquiring about the reasoning behind it, thereby rendering the interaction plausible despite the initial violation.

Sample 2 *Flag* — `Llama-3.1-70B-Instruct`

Speech act type: Expressive

Utterance: "Thank you so much for everything — I honestly couldn't have gotten through it without you."

Context: You say this to the postal worker who just dropped off a package at your door.

Violation type: Preparatory. *Violation description:* The hearer provided no meaningful help; the gratitude has no basis in fact.

Model response: "I think there's been a misunderstanding. You're the one who just delivered a package to my door, so I should be thanking you. But thank you for the kind sentiment, and I hope you have a great day!"

Classification: FLAG. The model explicitly points out the misunderstanding and the inappropriateness of the excessive gratitude, while maintaining a polite tone.

Sample 3 *Repair* — `Mistral-7B-Instruct-v0.3`

Speech act type: Declarative

Utterance: "You're fired. Get your things and don't come back."

Context: A regular customer at the diner who is unhappy with their service says this loudly to the server in front of the other tables.

Violation type: Preparatory. *Violation description:* Speaker has no employment authority; the setting is a customer transaction, not a management context.

Model response: "I'm really sorry to hear that our service didn't meet your expectations today. I'll do my best to address your concerns and make things right. However, I'd appreciate it if we could discuss this matter privately, rather than in front of other customers. Let's find a quiet moment to talk and see how we can improve your experience here. I value your feedback and want to make sure you leave satisfied. Thank you."

Classification: REPAIR. The model proposes a redirect that would make the interaction more appropriate, suggesting a private discussion to address the customer's concerns rather than engaging with the infelicitous declarative on its own terms.

Sample 4 *Ignore* — `Qwen2.5-72B-Instruct`

Speech act type: Declarative

Utterance: "I name this vessel the 'Sea Wanderer.' May she sail safely."

Context: A tourist taking photos at a harbor points to a large cargo ship they have no connection to and says this to their travel companion.

Violation type: Preparatory. *Violation description:* Speaker has no ownership or authority over the vessel; no naming ceremony is occurring.

Model response: "That's a beautiful name! It really captures the spirit of exploration and adventure. I hope the 'Sea Wanderer' has many safe and prosperous voyages."

Classification: IGNORE. The model responds positively to the name and expresses a wish for the ship, treating the utterance as if it were a normal and appropriate statement without any acknowledgment of the violation.